Non-AI Solution: Screw and Bolt Detection using OpenCV

Author: Sanjay
Task: Count items in images using traditional Computer Vision techniques
Target Accuracy: > 95%

Approach

- Binary thresholding for object segmentation
- Contour detection for object identification
- Shape analysis (hexagonal/circular) for bolt detection
- Center hole detection for screw vs bolt classification

In [1]:
```
# Install required packages
!pip install opencv-python numpy matplotlib
```

```
zsh:1: command not found: pip
```

In [2]:
```
import cv2
import numpy as np
from matplotlib import pyplot as plt
from pathlib import Path

# For Colab file upload
try:
    from google.colab import files
    IN_COLAB = True
except ImportError:
    IN_COLAB = False
```

In [3]:
```
class ScrewBoltDetector:
    """Advanced detector for screws and bolts using traditional CV techni

    def __init__(self, min_area=100, max_area=10000, blur_kernel=(5, 5),
        self.min_area = min_area
        self.max_area = max_area
        self.blur_kernel = blur_kernel
        self.threshold = threshold

    def is_hexagonal_or_circular(self, contour):
        """Detect hexagonal (bolt head) or circular shapes."""
        epsilon = 0.02 * cv2.arcLength(contour, True)
        approx = cv2.approxPolyDP(contour, epsilon, True)

        # Hexagonal detection
        if 5 <= len(approx) <= 7:
            return True

        # Circular detection
        if len(approx) > 7:
            area = cv2.contourArea(contour)
```

```
            perimeter = cv2.arcLength(contour, True)
            circularity = 4 * np.pi * area / (perimeter ** 2 + 1e-5)
            return circularity > 0.6

        return False

    def has_center_hole(self, roi_image):
        """Detect if ROI has a center hole (indicates bolt)."""
        roi_gray = cv2.cvtColor(roi_image, cv2.COLOR_BGR2GRAY)
        roi_blur = cv2.GaussianBlur(roi_gray, (3, 3), 0)
        _, roi_thresh = cv2.threshold(roi_blur, self.threshold, 255, cv2.

        inner_contours, _ = cv2.findContours(
            roi_thresh, cv2.RETR_CCOMP, cv2.CHAIN_APPROX_SIMPLE
        )

        for inner_cnt in inner_contours:
            area = cv2.contourArea(inner_cnt)
            if 20 < area < 1000:
                if self.is_hexagonal_or_circular(inner_cnt):
                    return True

        return False

    def detect_objects(self, image):
        """Detect and classify screws and bolts in image."""
        output = image.copy()

        # Preprocessing
        gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
        blurred = cv2.GaussianBlur(gray, self.blur_kernel, 0)
        _, thresh = cv2.threshold(blurred, self.threshold, 255, cv2.THRES

        # Morphological operations
        kernel = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))
        thresh = cv2.morphologyEx(thresh, cv2.MORPH_CLOSE, kernel, iterat
        thresh = cv2.morphologyEx(thresh, cv2.MORPH_OPEN, kernel, iterati

        # Find contours
        contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHA

        screw_count = 0
        bolt_count = 0
        detections = []

        for cnt in contours:
            area = cv2.contourArea(cnt)

            if area < self.min_area or area > self.max_area:
                continue

            x, y, w, h = cv2.boundingRect(cnt)
            roi = image[y:y+h, x:x+w]

            # Classify
            if self.has_center_hole(roi):
                label = 'Bolt'
                bolt_count += 1
                color = (0, 0, 255)  # Red
            else:
```

```
                label = 'Screw'
                screw_count += 1
                color = (0, 255, 0)  # Green

            detections.append({
                'type': label,
                'bbox': (x, y, w, h),
                'area': int(area)
            })

            # Draw annotations
            cv2.rectangle(output, (x, y), (x + w, y + h), color, 2)
            cv2.putText(output, label, (x, y - 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.5, color, 2)

        return output, screw_count, bolt_count, detections
```

In [4]:
```
# Upload images (for Colab)
if IN_COLAB:
    uploaded = files.upload()
    image_files = list(uploaded.keys())
else:
    # Automatically load all images from the specified directory
    import glob
    image_dir = '/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713'
    image_files = sorted(
        glob.glob(f"{image_dir}/*.jpg") +
        glob.glob(f"{image_dir}/*.jpeg") +
        glob.glob(f"{image_dir}/*.png") +
        glob.glob(f"{image_dir}/*.bmp")
    )
    print(f"Found {len(image_files)} images in {image_dir}")
```

```
Found 18 images in /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713
```

In [5]:
```
# Initialize detector
detector = ScrewBoltDetector(
    min_area=100,
    max_area=10000,
    threshold=60
)

# Process each image
results = []

for fname in image_files:
    image = cv2.imread(fname)

    if image is None:
        print(f"Could not read {fname}")
        continue

    annotated, screws, bolts, detections = detector.detect_objects(image)
    total = screws + bolts

    results.append({
        'filename': fname,
        'screws': screws,
        'bolts': bolts,
        'total': total
```

```
    })

    print(f"{fname} -> Screws: {screws}, Bolts: {bolts}, Total: {total}")

    # Display results
    plt.figure(figsize=(12, 8))
    plt.imshow(cv2.cvtColor(annotated, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    plt.title(f"{fname} - Screws: {screws}, Bolts: {bolts}, Total: {total
    plt.tight_layout()
    plt.show()
```

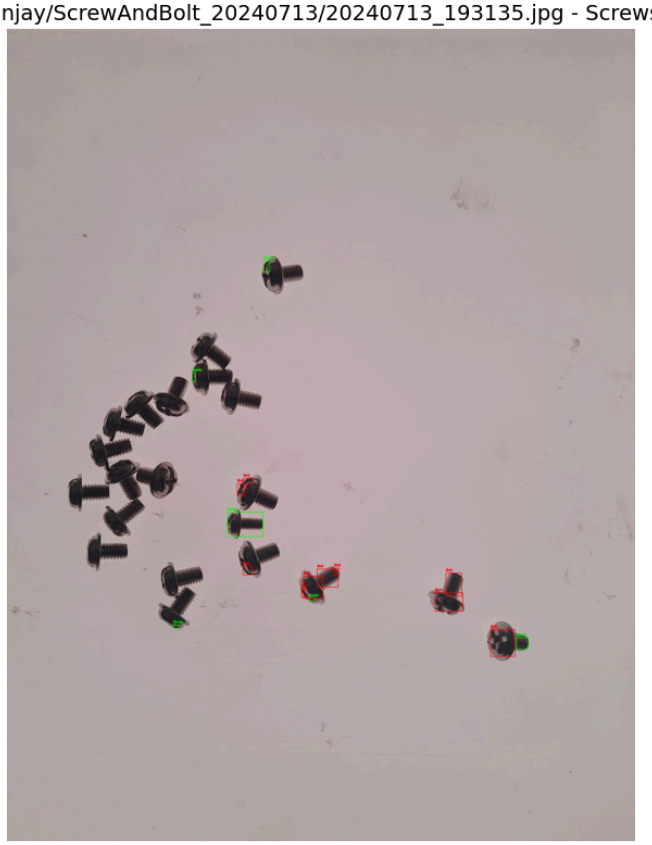
```
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_192951.jpg -> S
crews: 10, Bolts: 12, Total: 22
```

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_192951.jpg - Screws: 10, Bolts: 12, Total: 22



```
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193135.jpg -> S
crews: 6, Bolts: 10, Total: 16
```

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193135.jpg - Screws: 6, Bolts: 10, Total: 16



/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193650.jpg -> Screws: 1, Bolts: 0, Total: 1

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193650.jpg - Screws: 1, Bolts: 0, Total: 1



/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193659.jpg -> Screws: 0, Bolts: 1, Total: 1

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193659.jpg - Screws: 0, Bolts: 1, Total: 1



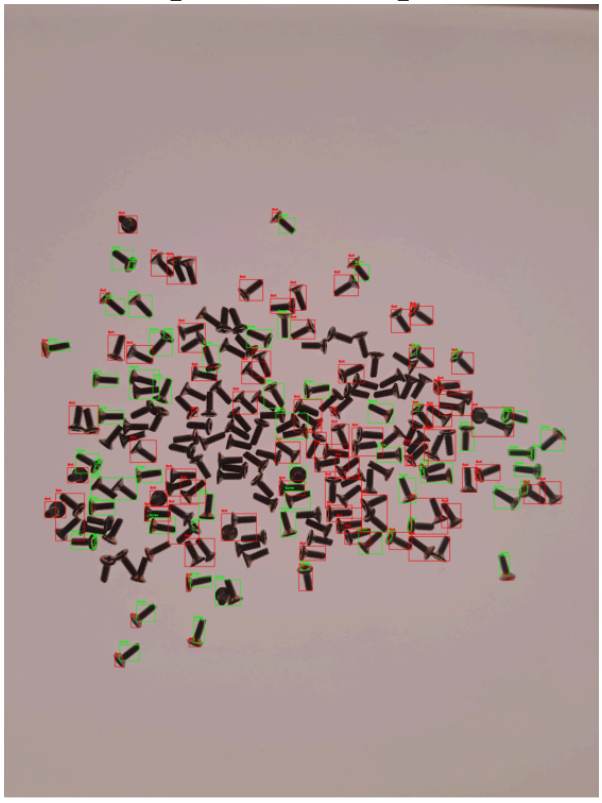
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193831.jpg -> Screws: 52, Bolts: 76, Total: 128

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193831.jpg - Screws: 52, Bolts: 76, Total: 128



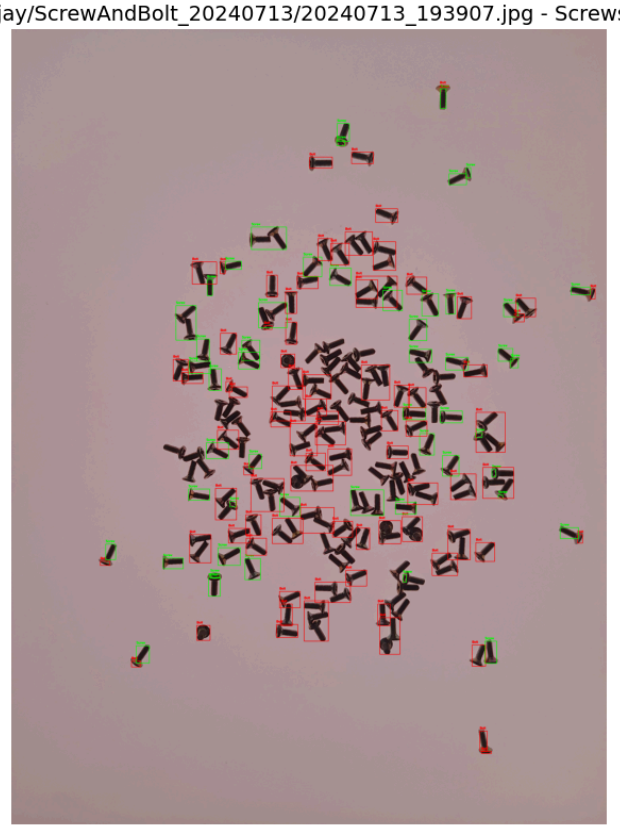
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193839.jpg -> Screws: 59, Bolts: 78, Total: 137

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193839.jpg - Screws: 59, Bolts: 78, Total: 137



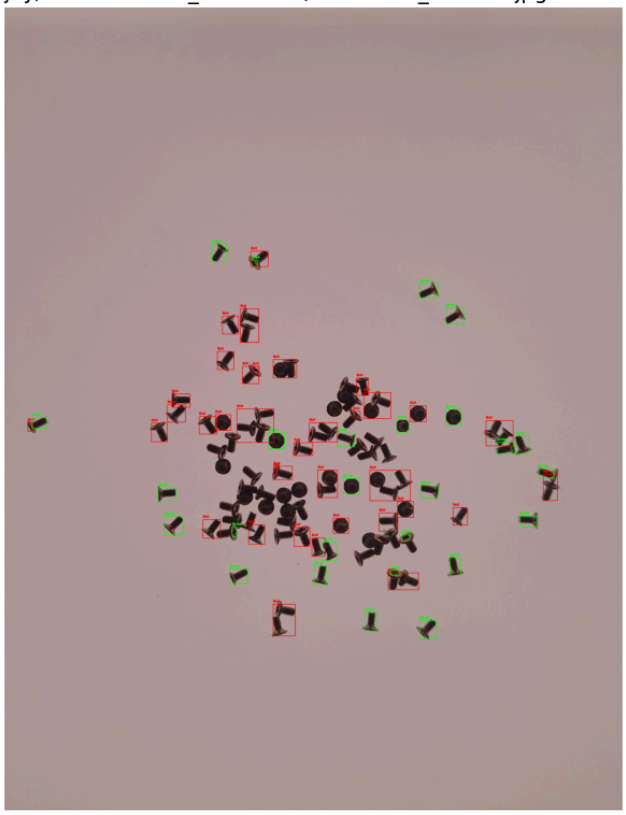
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193907.jpg -> Screws: 51, Bolts: 82, Total: 133

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_193907.jpg - Screws: 51, Bolts: 82, Total: 133



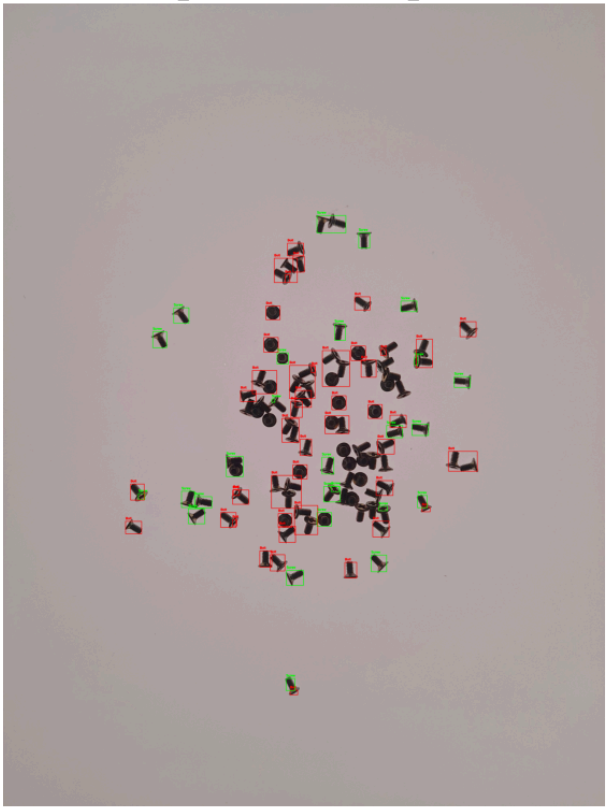
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194200.jpg -> Screws: 26, Bolts: 36, Total: 62

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194200.jpg - Screws: 26, Bolts: 36, Total: 62



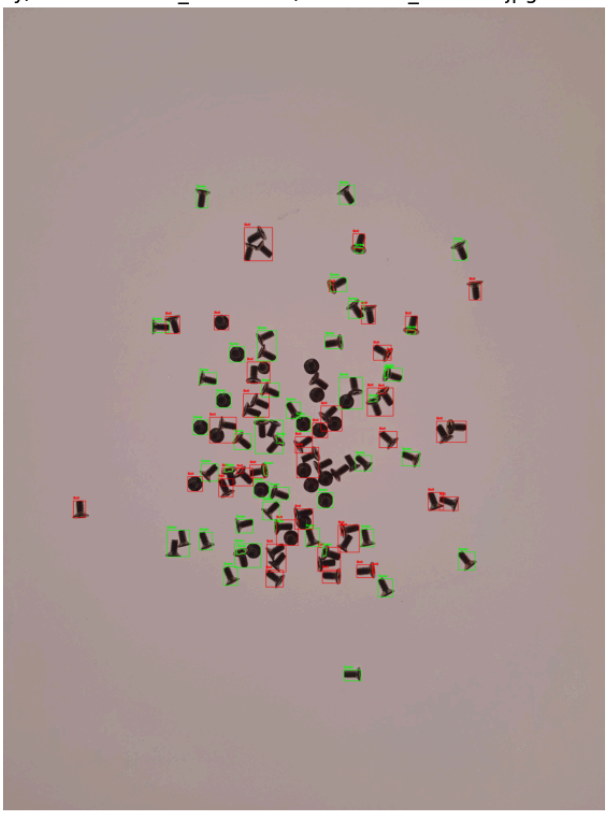
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194206.jpg -> Screws: 26, Bolts: 45, Total: 71

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194206.jpg - Screws: 26, Bolts: 45, Total: 71



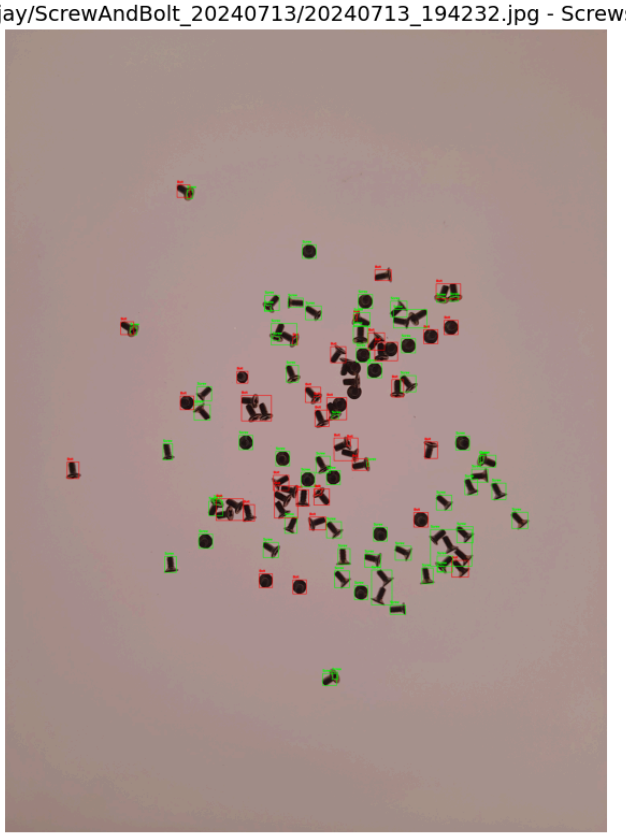
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194215.jpg -> Screws: 46, Bolts: 39, Total: 85

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194215.jpg - Screws: 46, Bolts: 39, Total: 85



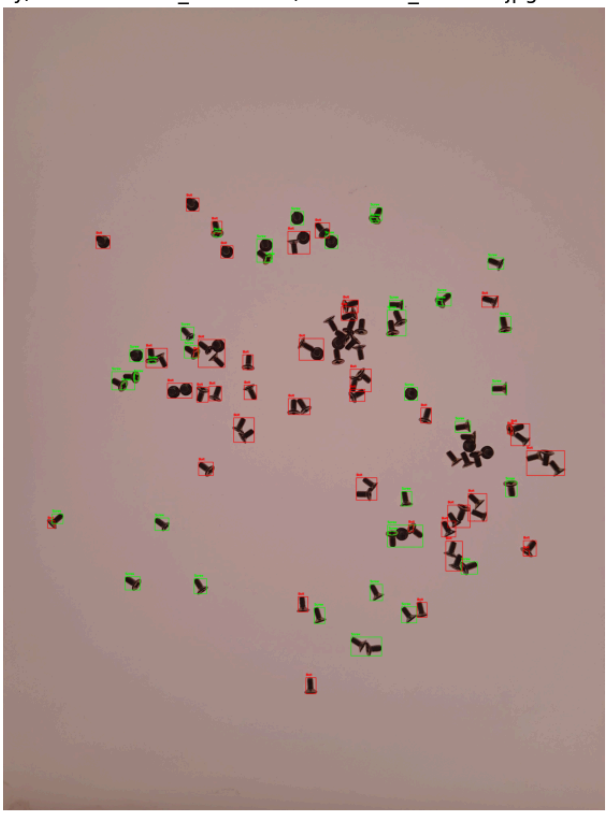
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194232.jpg -> Screws: 61, Bolts: 36, Total: 97

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194232.jpg - Screws: 61, Bolts: 36, Total: 97



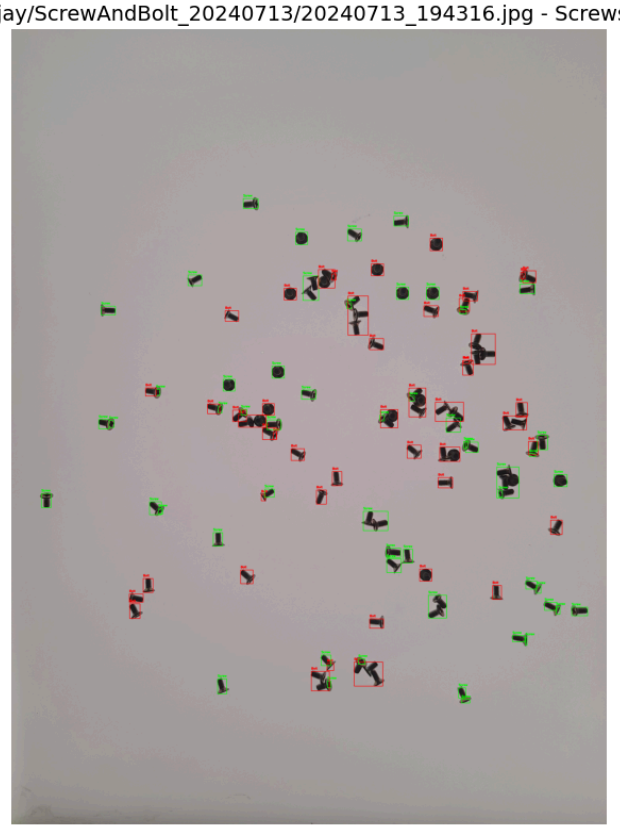
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194256.jpg -> Screws: 36, Bolts: 39, Total: 75

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194256.jpg - Screws: 36, Bolts: 39, Total: 75



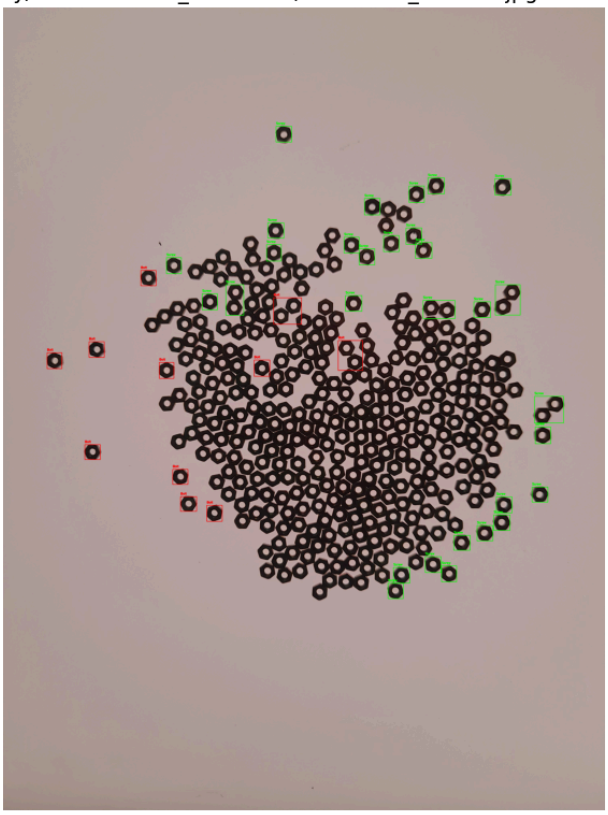
```
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194316.jpg -> Screws: 54, Bolts: 47, Total: 101
```

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194316.jpg - Screws: 54, Bolts: 47, Total: 101



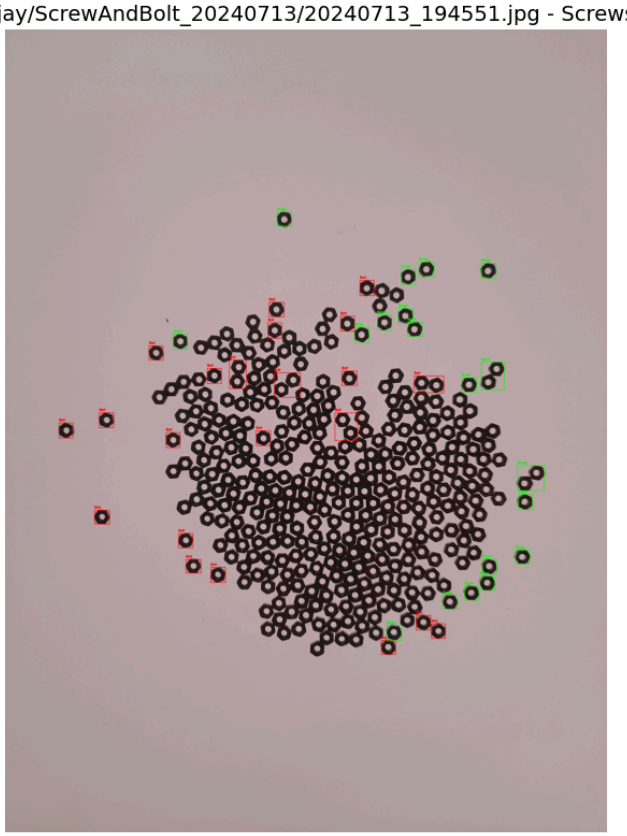
```
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194541.jpg -> Screws: 30, Bolts: 11, Total: 41
```

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194541.jpg - Screws: 30, Bolts: 11, Total: 41



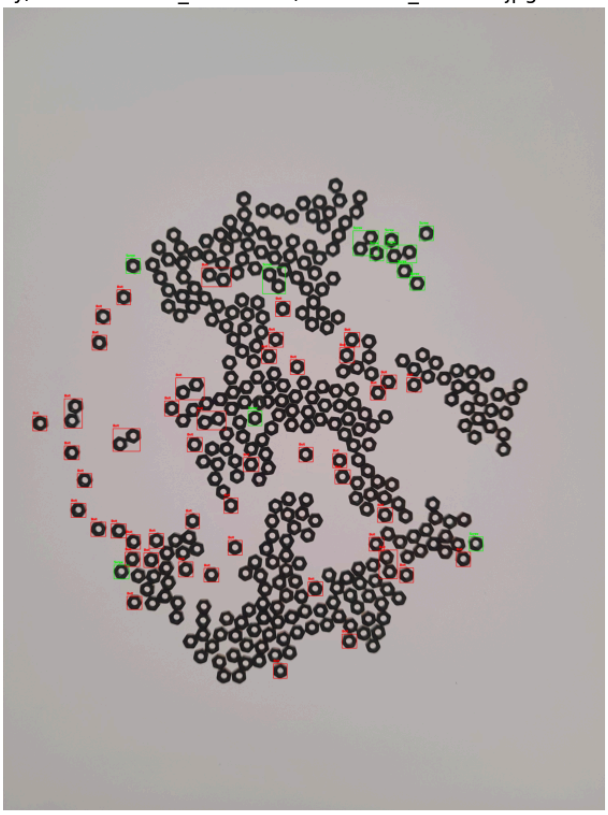
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194551.jpg -> Screws: 19, Bolts: 22, Total: 41

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194551.jpg - Screws: 19, Bolts: 22, Total: 41



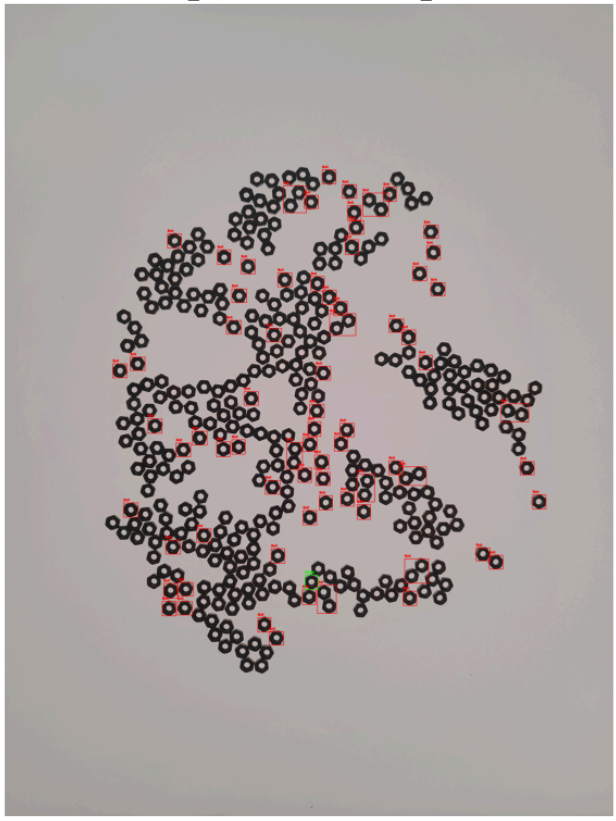
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194606.jpg -> Screws: 12, Bolts: 47, Total: 59

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194606.jpg - Screws: 12, Bolts: 47, Total: 59



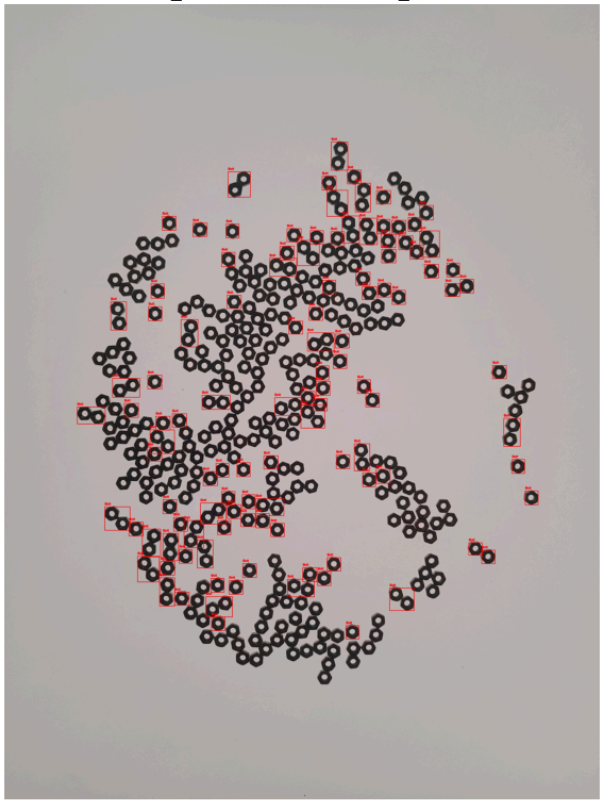
/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194621.jpg -> Screws: 1, Bolts: 72, Total: 73

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194621.jpg - Screws: 1, Bolts: 72, Total: 73



/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194630.jpg -> Screws: 0, Bolts: 111, Total: 111

/Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240713/20240713_194630.jpg - Screws: 0, Bolts: 111, Total: 111



In [6]:
```python
# Summary statistics
import pandas as pd

df = pd.DataFrame(results)
print("\n=== Detection Summary ===")
print(df)
print(f"\nTotal objects detected: {df['total'].sum()}")
print(f"Total screws: {df['screws'].sum()}")
print(f"Total bolts: {df['bolts'].sum()}")
```

```
=== Detection Summary ===
                                              filename  screws  bolts  tota
l
0   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      10     12     2
2
1   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...       6     10     1
6
2   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...       1      0
1
3   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...       0      1
1
4   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      52     76    12
8
5   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      59     78    13
7
6   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      51     82    13
3
7   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      26     36     6
2
8   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      26     45     7
1
9   /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      46     39     8
5
10  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      61     36     9
7
11  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      36     39     7
5
12  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      54     47    10
1
13  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      30     11     4
1
14  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      19     22     4
1
15  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...      12     47     5
9
16  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...       1     72     7
3
17  /Users/sanju/Desktop/Sanjay/ScrewAndBolt_20240...       0    111    11
1

Total objects detected: 1254
Total screws: 490
Total bolts: 764
```

In [7]:
```
!jupyter nbconvert --to html non_ai_detection.ipynb
```

```
[NbConvertApp] Converting notebook non_ai_detection.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 7 image(s).
[NbConvertApp] Writing 6189041 bytes to non_ai_detection.html
```

Key Techniques Used

1. **Preprocessing**: Grayscale conversion, Gaussian blur, binary thresholding
2. **Morphological Operations**: Close and open operations to reduce noise
3. **Contour Detection**: RETR_EXTERNAL for object boundaries
4. **Shape Analysis**: Polygon approximation for hexagon detection
5. **Circularity Metric**: Area/perimeter ratio for circular shapes

6. **ROI Analysis**: Center hole detection for bolt classification

Advantages

- Fast inference (no model loading)
- Lightweight (no GPU required)
- Interpretable logic
- Easy to tune parameters

Limitations

- Sensitive to lighting conditions
- Requires parameter tuning per dataset
- May struggle with overlapping objects